ROYAL CANADIAN NAVY

High Voltage Safety System of Work (SSOW) Authorized Person

QUALIFICATION STANDARD AND PLAN (QSP)

Naval Technical Officer (NTO) (00345),
Marine Technician Electrician (MAR TECH) (00379-01, 00379-03)

Common Specialty Qualification - (AMBX)

MITE Course IDs:
Authorized Person (AP) – DL 123126 & OJPR 123127

RECORD OF AMENDMENTS

This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear at the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History.”

Revision	Approval Date	Authority
1.0 (QSPWB March 2021)	4 July 2023	CNTG

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FOREWORD

This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

A-P9 - 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

A-PD-055-002/PP-003 Canadian Forces Officer General Specification;

A-PD-055-002/PP-002 Non-Commissioned Member General Specification;

A-PD-055-002/PP-001 Name of Occupation Job Based Specification or Job-Based Occupational Specification;

A-PD-055-003/PQ-001 Name of Specialty Specification; and

NAVORD 4500-0 Structured Delegation; and

NPTORD 4500-22 Qualification Standard and Plan Production and Management.

This QSP consists of specific Performance Objectives (POs), Activity and Assessment Plan (AAP) in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

This QSP is effective upon approval and supersedes previous versions.

Note: HV AOPV training is a new capability, and this is the first QSP version.

Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NFS(P)), who will forward them to NPTG for approval, as appropriate.

The designated TEs for this training at this time is not applicable. The HV interim training consists of eLearning via distributed learning and OJPR on ship.

TE	Phase / Module	MITE Course IDs	NQual
NFS(A/P)	DL	123126	AMBX
HMC Ship	OJPR	123127	AMBX

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

MARPAC RDIMS # 1381603
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the **Authorized Person (AP)**. This QSP is designed to assist the training system in developing, conducting, evaluating, and validating the training.

Authorized Person: AP qualified personnel are Sailor 1st Class and above Marine Technicians – Electricians who are fully conversant with HV systems and possess in-depth knowledge of electrical safety rules, and have undergone relevant HV training, which allows them to act as an Authorized Person in Control (APC). The APC is the only person authorized to isolate HV equipment in accordance with the HV ship-class SEMS. AP qualified personnel may conduct work on live or dead HV equipment.

Interim State Training Overview:

As part of the interim state training, the HV Individual Training & Education (IT&E) will be delivered through various delivery models based on the HV qualification:

- AP: eLearning followed by OJPR on ship

COURSE DURATION

2. Course duration is calculated following NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP. Note that the DL timings are approximations and must be validated during the post pilot course review.

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points Authorized Person	AP: Unlimited (online)
Performance Checks (PCs) Authorized Person	145 mins per trainee + 5 min debrief
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	N/A
Training Administrative Time (TAT)	N/A
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	N/A
Sub-total	N/A
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days (rounded up)**	
Total Training Days	N/A

RESOURCES

3. The following table details key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
OJPR Trainer	PO1	Sr. Electrical Propulsion Mgr AE and AP Qualifications	1	PO 001
OJPR Trainer	Lt(N)	Naval Technical Officer AE and AP Qualifications	1	PO 001

4. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP.

Critical Training Resources	PO/EO
1. DND computer tablets (only if the trainee does not have a personal computer at home to conduct DL) 2. AOPV ship 3. DLN 3.0 4. OJPR Package	PO 001

COURSE CAPACITY

5. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (OJPR coach to Learner Ratios). Justification can be found in Annex C.

Distributed Learning for AP members

Note: given the eLearning courses are not facilitated and conducted via distributed learning, there are no requirements for max, min, or optimum.

OJPR on AOPV

	CP	**AP**	**AE**
Max	6	6	6

Min.	1	1	1
Optimum	4	4	4

Notes:

1. Separate OJPRs will be conducted based on HV qualification
2. The following trainee numbers are based on the new Arctic Offshore Patrol Vessell (AOPV) Naval Technical Department team:

- AE: 2 (includes Sr. Electrical Propulsion Manager (SEPM))
- AP: 8 (include MSE Naval Technical Officer (NTO) who must complete AP qualification)
- CP: 15

MITE

6. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System (NTS). It is scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

8. Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

Distributed Learning:

AP Qualification Course

RQ-S1 for MAR Tech Elecs

MSE HOD for NTO

HV Safety Instruction (Calhoon MEBA, or equivalent)

OJPR:

AP

AP eLearning Course

9. Qualification. Successful completion of this course will grant the trainee the national AMBX **High Voltage Training** qualification.

COURSE REPORTING

10. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition,” DAOD 5031-1 “Canadian Forces Military Equivalencies Program,” and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.”

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

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GENERAL

12. This is a pass/fail course. Trainees must successfully complete only the POs related to their occupation IAW NAVORD 11100-1 on ship represented in the following table:

Qualification	AP
Performance Objective	PO 001
Occupation	• MAR Tech Elec S1 and above (Technician) • NTO

13. Performance Checks (PCs) are based on the Performance Objectives (POs) in Chapter 3 of this QSP. Only the PCs related to the trainee's designated HV role must be successfully completed to achieve the qualification.

14. Trainees will be developed and assessed using a combination of OJPR and summative assessment events. See Annex K for details.

15. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct, which may result in the convening of a Training Review Board (TRB).

PERFORMANCE CHECKS

16. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the PO conditions (where possible) and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

17. PCs will be conducted after the trainees have successfully completed their OJPR package, demonstrating that they have had the opportunity to develop skills and apply knowledge related to the PO.

18. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process.” Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

19. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners.”

20. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

21. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning so that content is retained.

22. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

23. Continuous monitoring of candidate progress is required to provide the following:

- Early warning of trainee difficulties;
- Feedback on the effectiveness of training;
- Information for any required TRBs; and
- Information to assist in the completion of the CF 377 Course Report.

- Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.
- Instructional staff shall maintain individual trainee files, including the results of all:
- Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);
- Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and
- Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

24. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process.”

25. In the event of prolonged trainee absence, TE Staff shall determine, on a case-by-case basis, the necessity to convene a TRB.

COURSE REPORTS

26. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications.”

27. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

28. The Course Report narrative comments could include:
- Ability to work with others;
- Ability to work under pressure;
- Written and verbal communication skills;
- Organizational skills; and
- Leadership ability.

- Training limitations and/or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

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PO 001 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Perform High Voltage Safe System of Work Procedures as an Authorized Person.

2. **Conditions:**

   a. **Given:**

      (1) Supervision;

      (2) Safety Equipment;

      (3) HV Tools;

      (4) Assistance;

      (5) PPE; and

      (6) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Day or night;

      (2) At sea or alongside; and

      (3) HV live/dead.

3. **Standard:** The member shall perform high voltage safe system of work procedures as an authorized person (AP) to include:

   a. Completing the high voltge SSOW administration procedural roles;

   b. Supervising a permit to work (PTW) on de-energized or energized HV equipment as a PiC AP;

   c. Supervising a sanction for test (SFT) as a PiC AP;

   d. Performing HV Switching Program (HVSP) preparation work as an APC;

   e. Performing an HV equipment isolation procedure as an APC IAW the pre-approved HVSS;

f. Performing a PTW procedure as an APC;

g. Performing an SFT procedure as an APC;

h. Performing a Limitation of Access (LOA) procedure as an APC; and

i. Performing an HV restoration procedure as an APC IAW the pre-approved HVSS.

4. **References:** A1, A2, A3, A4, A5, A6, C1, C2

5. **Task Numbers:** A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16.

6. **Remarks:**

a. There are no physical trainers, and the training schedule is limited due to the availability of instructional HV SMEs and AOPV ship.

EO 001.01 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Supervising a PTW on de-energized equipmentas a PiC AP.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil;

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall supervise a PTW on de-energized equipment as a PiC AP IAW to include:

   a. Completing the HVSSOW procedures as an AP;

   b. Physically reporting to Authorized Person in Control (APC) for verification to include:

      (1) High voltage work credentials IAW SOP-S13;

      (2) Worksite PPE is secured with PiC to include:

         (a) Hearing protection;

         (b) Eye protection;

         (c) Steel toe boots; and

         (d) NCD;

      (3) Personal key safe padlock is secured with PiC;

      (4) Personal key safe key(s) is secured with PiC; and

      (5) Padlock identification tag(s) is secured with PiC;

c. Signing High Voltage Switching Schedule (HVSS) form IAW Annex SOP-S13B to accept equipment isolation after physical HV system walkthrough by APC to include:

(1) Isolation points;

(2) Earthing points;

(3) HVSP devices used; and

(4) Safety brief at POW;

d. Initialling part 1 of the PTW form IAW Annex SOP-S13C to accept any special instructions and safety measures;

e. Signing part 2 of the PTW form IAW Annex SOP-S13C to accept equipment isolation by APC to include:

(1) Reading part 2 of the PTW form; and

(2) Visually verifying APC places bare hand on all generator connections to prove HV circuit is dead;

f. Accessing switchboard key safe to include:

(1) Placing key safe padlock on the switchboard key safe multi-lock hasp before HV work commences; and

(2) Removing key safe padlock from switchboard key multi-lock hasp upon completion of HV work;

g. Supervising HV work performed by external contractors to include:

(1) Work being completed IAW PTW; and

(2) Safety measures being followed IAW HV SSOW;

h. Visually verifying the work area is free of hazards upon completion of work to include clearing:

(1) Personnel;

(2) Tools;

(3) Instruments;

(4) Temporary earths; and

(5) Foreign objects;

i. Verbally informing the APC that the PTW is complete to include:

(1) Completing part 3 of the PTW form; and

(2) Returning parts 1 and 2 of the PTW form to the APC.

4. Time Required:

a. EO – 4 x 50 min periods.

5. Instructional Methods/Media:

a. Online instruction (DLN 3.0).

6. Training Aids/Resources Required:

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. Test Details:

a. EO 001.01 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. References: A1, A2, A3, A4, A5, A6, C1, C2

9. Remarks:

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. Teaching Points:

Lesson Objective: Supervising a PTW on de-energized equipment as a PiC AP			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Physically reporting to Authorized Person in Control (APC) for verification to include:	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2

a) High voltage work credentials IAW SOP-S13; b) Worksite PPE is secured with PiC to include: i. Hearing protection; ii. Eye protection; iii. Steel toe boots; and iv. NCD. c) Personal key safe padlock is secured with PiC; d) Personal key safe key(s) is secured with PiC; and e) Padlock identification tag(s) is secured with PiC.			
Signing High Voltage Switching Schedule (HVSS) form IAW Annex SOP-S13B to accept equipment isolation after physical HV system walkthrough by APC to include: a) Isolation points; b) Earthing points; c) HVSP devices used; and d) Safety brief at POW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Forms Annex A – Annex Z
Initialling part 1 of the PTW form IAW Annex SOP-S13C to accept any special instructions and safety measures;	Direct instruction	Online delivery via DLN 3.0	A1, A2, PTW Form

Signing part 2 of the PTW form IAW Annex SOP-S13C to accept equipment isolation by APC to include: a) Reading part 2 of the PTW form; and b) Visually verifying APC places bare hand on all generator connections to prove HV circuit is dead.	Direct instruction	Online delivery via DLN 3.0	A1, A2, PTW Form
Accessing switchboard key safe to include: a) Placing key safe padlock on the switchboard key safe multi-lock hasp before HV work commences; and b) Removing key safe padlock from switchboard key multi-lock hasp upon completion of HV work.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A6, C1, C2 PTW Form
Supervising HV work performed by external contractors to include: a) Work being completed IAW PTW; and b) Safety measures being followed IAW HV SSOW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A5, A6, C1, C2 PTW Form
Visually verifying the work area is free of hazards upon completion of work to include clearing: a) Personnel;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 PTW Form

b) Tools; c) Instruments; d) Temporary earths; and e) Foreign objects.			
Verbally informing the APC that the PTW is complete to include: a) Completing part 3 of the PTW form; and b) Returning parts 1 and 2 of the PTW form to the APC.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 PTW Form

- EO 001.01 shall be assessed in PC 001

EO 001.02 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Supervising a Sanction for Test (SFT) as a PiC AP.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall supervise a Sanction for Test (SFT) as a PiC AP IAW to include:

   a. Physically reporting to Authorized Person in Control (APC) for verification to include:
      (1) High voltage work credentials IAW SOP-S13;
      (2) Worksite PPE is secured with PiC to include:
         (a) Hearing protection;
         (b) Eye protection;
         (c) Steel toe boots; and
         (d) NCD;
      (3) Personal key safe padlock is secured with PiC;
      (4) Personal key safe key(s) is secured with PiC; and
      (5) Padlock identification tag(s) is secured with PiC;

   b. Signing the HVSS form IAW Annex SOP-S13B to accept the HV equipment isolation after physical HV system walkthrough by APC to include:
      (1) Isolation points;
      (2) Earthing points;

(3) HVSP devices used; and

(4) Safety brief at POW;

c. Initialling part 1 of the SFT form IAW Annex SOP-S13D to accept any special instructions and safety measures;

d. Signing part 2 of the SFT form IAW Annex SOP-S13D to accept the HV equipment isolation by APC to include:

(1) Reading part 2 of the SFT form; and

(2) Visually verifying the APC places bare hand on all generator connections to prove the HV circuit is dead;

e. Accessing the HV switchboard key safe to include:

(1) Placing the key safe padlock on the switchboard key safe multi-lock hasp before HV work commences; and

(2) Removing the key safe padlock from switchboard key safe upon completion of HV work;

f. Supervising the HV work by external contractors to include:

(1) Work being completed IAW SFT; and

(2) Safety measures being followed IAW HV SSOW;

g. Visually verifying the work area is free of hazards upon completion of work to include clearing:

(1) Personnel;

(2) Tools;

(3) Instruments;

(4) Temporary earths; and

(5) Foreign objects;

h. Verbally informing the APC that the SFT is complete IAW Annex SOP-S13D to include:

(1) Completing part 3 SFT form; and

(2) Returning parts 1 and 2 SFT form to APC.

4. **Time Required:**

a. EO – 4 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.02 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References**: A1, A2, A3, A4, A5, A6, C1, C2,

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Supervising a Sanction for Test (SFT) as a PiC AP			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Physically reporting to Authorized Person in Control (APC) for verification to include: a) High voltage work credentials IAW SOP-S13; b) Worksite PPE is secured with PiC to include:	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2

i. Hearing protection; ii. Eye protection; iii. Steel toe boots; and iv. NCD. c) Personal key safe padlock is secured with PiC; d) Personal key safe key(s) is secured with PiC; and e) Padlock identification tag(s) is secured with PiC.			
Signing the HVSS form IAW Annex SOP-S13B to accept the HV equipment isolation after physical HV system walkthrough by APC to include: a) Isolation points; b) Earthing points; c) HVSP devices used; and d) Safety brief at POW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Scheule Form Annex A – Annex Z
Initialling part 1 of the SFT form IAW Annex SOP-S13D to accept any special instructions and safety measures;	Direct instruction	Online delivery via DLN 3.0	A1, A2, Sanction for Test Form
Signing part 2 of the SFT form IAW Annex SOP-S13D to accept the HV equipment isolation by APC to include: a) Reading part 2 of the SFT form; and	Direct instruction	Online delivery via DLN 3.0	A1, A2, Sanction for Test Form

b) Visually verifying the APC places bare hand on all generator connections to prove the HV circuit is dead.			
Accessing the HV switchboard key safe to include: a) Placing the key safe padlock on the switchboard key safe multi-lock hasp before HV work commences; and b) Removing the key safe padlock from the switchboard key safe upon completion of HV work.	Direct instruction	Online delivery via DLN 3.0	A1, A2, Sanction for Test Form
Supervising the HV work by external contractors to include: a) Work being completed IAW SFT; and b) Safety measures being followed IAW HV SSOW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2, Sanction for Test Form
Visually verifying the work area is free of hazards upon completion of work to include clearing: a) Personnel; b) Tools; c) Instruments; d) Temporary earths; and e) Foreign objects.	Direct instruction	Online delivery via DLN 3.0	A1, A2, Sanction for Test Form

Verbally informing the APC that the SFT is complete IAW Annex SOP-S13D to include: a) Completing part 3 SFT form; and b) Returning parts 1 and 2 SFT form to APC.	Direct instruction	Online delivery via DLN 3.0	A1, A2, Sanction for Test Form

- EO 001.02 shall be assessed in PC 001

EO 001.03 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Performing HV Switching Program (HVSP) preparation work as an APC.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil;

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform HV Switching Program (HVSP) preparation work as an APC IAW to include:

   a. Initiating the serial tracking list IAW Annex SOP-S13F to include:

      (1) Date issued;
      (2) Serial number;
      (3) Form type;
      (4) Purpose
      (5) System/equipment;
      (6) PiC; and
      (7) Key safe;

   b. Completing the HVSS form IAW the pre-approved HVSS form to include:

      (1) Making a duplicate copy of HVSS under the same serial number;

      (2) Using original copy to perform HVSS; and

      (3) Providing a duplicate copy to PiC;

   c. Obtaining AE initials under the isolation procedure step 1 in the HVSS form;

d. Conducting a verbal briefing to duty staff to include:

(1) Sr. Watch keeper and Duty Engineer while alongside; and

(2) Officer of Watch and Engineering Watch keeper at sea;

e. Updating the HV stateboards to include:

(1) Shipwide HV stateboard(s) while alongside; and

(2) Equipment HV stateboard while alongside;

f. Physically handing over an HV informative pipe script to ship's Quartermaster to include:

(1) Developing a scripted HV status pipe information to include:

(a) HV dead; and

(b) HV live.

4. **Time Required:**

a. EO – 4 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.03 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A4

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Performing HV Switching Program (HVSP) preparation work as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Initiating the serial tracking list IAW Annex SOP-S13F to include: a) Date issued; b) Serial number; c) Form type; d) Purpose e) System/equipment; f) PiC; and g) Key safe.	Direct instruction	Online delivery via DLN 3.0	A1, A2 High Voltage Switching Schedule Form, Permit to Work Form, Sanction for Test Form, Limiation of Access Form, Master Key Register, Form, APC Handover Form
Completing the HVSS form IAW the pre-approved HVSS form to include: a) Making duplicate copy of HVSS under same serial number; b) Using original copy to perform HVSS; and c) Providing duplicate copy to PiC.	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Form Annex A – Annex Z
Obtaining AE initials under the isolation procedure step 1 in the HVSS form	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Form Annex A – Annex Z
Conducting a verbal briefing to duty staff to include:	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4

a) Sr. Watch keeper and Duty Engineer while alongside; and b) Officer of Watch and Engineering Watch keeper at sea.			
Updating the HV stateboards to include: a) Shipwide HV stateboard(s) while alongside; and b) Equipment HV stateboard while alongside.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4
Physically handing over an HV informative pipe script to ship's Quartermaster to include: a) Developing a scripted HV status pipe information to include: b) HV dead; and c) HV live.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4

- EO 001.03 shall be assessed in PC 001

EO 001.04 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Performing an HV equipment isolation procedure as an APC.

2. **Conditions:**

   a. Given:

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform an HV equipment isolation procedure as an APC IAW the pre-approved HVSS to include:

   a. Initialling each isolation step in the HVSS form upon completion;

   b. Setting up a safe work area with restricted access to include:

      (1) Cordoning off area with tape; and

      (2) Affixing signage;

   c. Visually inspecting the min required arc flash PPE IAW arc flash rating by sighting affixed label on HV equipment to be isolated;

   d. Opening the breaker remotely;

   e. Recording the decay time;

   f. Racking out the breaker after applicable decay time;

   g. Applying a circuit main earth (CME);

   h. Placing all HVSS keys in the switchboard key safe;

   i. Donning arc flash PPE;

   j. Proving HV circuit is dead by:

(1) Obtaining a CP qualified or above personnel to act as a safety member to include:

(a) Briefing the member with specific instructions in case of an incident; and

(b) Providing a safety cane to the safety member;

(2) Testing connections with the Potential Indicator (PI) to include:

(a) Phase to phase; and

(b) Phase to ground;

(3) Visually checking all generator connections for HV; and

(4) Re-testing the PI;

k. Obtaining an AP qualified member to visually check the completed equipment isolation procedure IAW the pre-approved switching schedule; and

l. Placing the completed HVSS form in the switchboard register.

4. **Time Required:**

a. EO – 3 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.04 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4, A5, A6, A7,C1, C2

9. **Remarks:**

   a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Performing an HV equipment isolation procedure as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Initialling each isolation step in the HVSS form upon completion	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Form Annex A- Annex Z
Setting up a safe work area with restricted access to include: a) Cordoning off area with tape; and b) Affixing signage.	Direct instruction	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Form Annex A- Annex Z
Visually inspecting the min. required arc flash PPE IAW arc flash rating by sighting affixed label on HV equipment to be isolated	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2 High Voltage Switching Schedule Form Annex A- Annex Z
Opening the breaker remotely	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A7, High Voltage Switching Schedule Form Annex A- Annex Z
Recording the decay time;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A7, High Voltage Switching Schedule Form Annex A- Annex Z

Racking out the breaker after applicable decay time HV live.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form
Applying a circuit main earth	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form
Placing all HVSS keys in the switchboard key safe	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form
Donning arc flash PPE	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form
Proving HV circuit is dead by: a) Obtaining a CP qualified or above personnel to act as a safety member to include: b) Briefing the member with specific instructions in case of an incident; and c) Providing a safety cane to the safety member. d) Testing connections with the Potential Indicator (PI) to include: e) Phase to phase; and f) Phase to ground. g) Visually checking all generator connections for HV; and h) Re-testing the PI.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form
Obtaining an AP qualified member to check the completed equipment isolation procedure visually IAW	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form

the pre-approved switching schedule			
Placing the completed HVSS form in the switchboard register	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form

- EO 001.04 shall be assessed in PC 001

EO 001.05 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Performing a PTW procedure as an APC.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil;

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform a PTW procedure as an APC to include:

   a. Preparing for a PTW to include:

      (1) Completing part 1 of the PTW form;

      (2) Documenting special instructions and safety measures in part 1 of the PTW form if required; and

      (3) Obtaining AE signature on part 1 of the PTW form;

   b. Issuing a PTW IAW Annex SOP-S13C by:

      (1) Visually verifying the PiC to include:

         (a) HV work credentials;

         (b) Personal key safe padlock is secured with PiC;

         (c) Personal key safe key(s) is secured with PiC; and

         (d) Padlock identification tag(s) is secured with PiC;

      (2) Conducting an HV system walkthrough with the PiC at POW to include:

         (a) Isolation points;

(b) Earthing points;

(c) HVSP devices used; and

(d) Safety brief at POW;

(3) Physically proving the HV circuit is dead to the PiC by:

(a) Placing bare hand on all generator connections; and

(b) Verbally asking PiC if there is anything else to prove dead;

(4) Obtaining PiC's initial in the HVSS form for the acceptance of HV equipment isolation;

(5) Obtaining PiC's initials part 1 of the PTW form IAW Annex SOP-S13C for acceptance of special instructions and safety measures if required;

(6) Obtaining PiC's signature part 2 of the PTW form IAW Annex SOP-S13C for the acceptance of HV equipment isolation;

(7) Visually inspecting PiC's PPE before HV work commences;

(8) Making a duplicate copy of the PTW under the same serial number to include:

(a) Placing the original copy in the switchboard register; and

(b) Providing a duplicate copy to the PiC;

c. Cancelling a PTW IAW Annex SOP-S13C by:

(1) Visually verifying the cancellation reason in part 3 of the PTW is complete; and

(2) Briefing AE the reason for cancelling the PTW;

d. Closing a PTW IAW Annex SOP-S13C by:

(1) Visually verifying the work area is free of hazards upon completion of work to include clearing:

(a) Personnel;

(b) Tools;

(c) Instruments;

(d) Temporary earths; and

(e) Foreign objects;

(2) Visually verifying PiC has completed part 3 of the PTW form;

(3) Completing part 4 of the PTW form to include:

(a) Signing both copies of part 4 of the PTW form; and

(b) Placing one completed PTW copy into the switchboard register;

(4) Completing serial tracking list IAW Annex SOP-S13F to include:

(a) Date completed; and

(b) Initials.

4. **Time Required:**

a. EO – 4 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.05 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4, A5, A6, C1, C2

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Performing a PTW procedure as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Preparing for a PTW to include: a) Completing part 1 of the PTW form; b) Documenting special instructions and safety measures in part 1 of the PTW form if required; and c) Obtaining AE signature on part 1 of the PTW form.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 & High Voltage Swtiching Schedule Form Annex A- Annex Z
Issuing a PTW IAW Annex SOP-S13C by: Visually verifying the PiC to include: a) HV work credentials; b) Personal key safe padlock is secured with PiC; c) Personal key safe key(s) is secured with PiC; and d) Padlock identification tag(s) is secured with PiC.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Conducting an HV system walkthrough with the PiC at POW to include: a) Isolation points; b) Earthing points; c) HVSP devices used; and d) Safety brief at POW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Physically proving the HV circuit is dead to the PiC by:	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form

a) Placing bare hand on all generator connections; and b) Verbally asking PiC if there is anything else to prove dead.			Annex A- Annex Z
Obtaining PiC's initial in the HVSS form for the acceptance of HV equipment isolation	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Obtaining PiC's initials part 1 of the PTW form IAW Annex SOP-S13C for acceptance of special instructions and safety measures if required	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Obtaining PiC's signature part 2 of the PTW form IAW Annex SOP-S13C for the acceptance of HV equipment isolation	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Visually inspecting PiC's PPE before HV work commences	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, High Voltage Switching Schedule Form Annex A- Annex Z
Making a duplicate copy of the PTW under the same serial number to include: a) Placing the original copy in the switchboard register; and b) Providing a duplicate copy to the PiC	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Permit to work Form
Cancelling a PTW IAW Annex SOP-S13C by:	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Permit to work Form

a) Visually verifying the cancellation reason in part 3 of the PTW is complete; and b) Briefing AE the reason for cancelling the PTW.			
Closing a PTW IAW Annex SOP-S13C by: a) Visually verifying the work area is free of hazards upon completion of work to include clearing: i. Personnel; ii. Tools; iii. Instruments; iv. Temporary earths; and v. Foreign objects. b) Visually verifying PiC has completed part 3 of the PTW form; and c) Completing part 4 of the PTW form to include: i. Signing both copies of part 4 of the PTW form; and ii. Placing one completed PTW copy into the switchboard register. d) Completing serial tracking list IAW Annex SOP-S13F to include: i. Date completed; and ii. Initials.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Permit to work Form

Performing a SFT procedure as an APC to include: a) Preparing for a SFT to include: i. Completing part 1 of the SFT form; ii. Documenting any special instructions and safety measures in part 1 SFT form; and iii. Obtaining the AE's signature on part 1 of the SFT form.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, Sanction for Test Form
Issuing a SFT IAW Annex SOP-S13D by: a) Visually verifying the work area is free of hazards upon completion of work to include clearing: i. Personnel; ii. Tools; iii. Instruments; iv. Temporary earths; and v. Foreign objects.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, Sanction for Test Form
Conducting an HV system walkthrough at POW to include: a) Isolation points; b) Earthing points; c) HVSP devices used; and d) Safety brief at POW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, Sanction for Test Form
Obtaining PiC's initial in the HVSS form for the acceptance of HV equipment isolation	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, Sanction for Test Form

Obtaining PiC's signature part 2 of the SFT form for the acceptance of HV equipment isolation	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, Sanction for Test Form
Visually inspecting PiC's PPE before HV work commences;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2, Sanction for Test Form
Making a duplicate copy of the SFT under the same serial number to include: a) Placing the original copy in switchboard register; and b) Providing a duplicate copy to the PiC	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Sanction for Test Form
Cancelling a SFT IAW Annex SOP-S13D by: a) Verifying the reason for cancelling the SFT in part 3 of the form; b) Briefing the AE the reason for cancelling the SFT;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Sanction for Test Form
Closing an SFT IAW Annex SOP-S13D by: a) Visually inspecting the work site to include: i. Clean work area; and ii. All tools, test equipment and materials removed from work area. b) Visually verifying the PiC AP has completed part 3 of the SFT form;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Sanction for Test Form

c) Completing part 4 of the SFT form to include: d) Signing both copies of part 4 of the SFT form; e) Placing one completed SFT copy into the switchboard register. f) Completing serial tracking list IAW Annex F SOP-S13 to include: i. Date completed; ii. Initials.			

- EO 001.05 shall be assessed in PC 001

EO 001.07 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Performing a Limitation of Access (LOA) procedure as an APC.

2. **Conditions:**

   a. Given:

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform a Limitation of Access (LOA) procedure as an APC IAW refs to include:

   a. Preparing for a LOA to include:

      (1) Completing serial tracking list IAW Annex SOP-S13F;

      (2) Completing part 1 LOA form; and

      (3) Documenting special instructions and safety measures in part 1 LOA form if required;

   b. Issuing a LOA IAW Annex SOP-S13E by:

      (1) Visually verifying PiC's credentials;

      (2) Conducting a worksite walkthrough with PiC at POW to include:

         (a) Extent of work to be carried out;

         (b) Live HV equipment in or adjacent to the area where work is to be conducted; and

         (c) Safety brief at POW;

      (3) Cordoning off the work area with caution tape;

(4) Obtaining the PiC's signature part 1 of the LOA form for acceptance of special instructions and safety measures if required;

(5) Obtaining the PiC's signature on part 2 of the LOA form for acceptance of scope and limitations of work; and

(6) Making a duplicate copy of the LOA form under the same serial number to include:

(a) Placing the original copy in the switchboard register; and

(b) Providing a duplicate copy to the PiC;

c. Inspecting the worksite at the end of each working period or day to include:

(1) Flammable or hazardous materials are removed;

(2) Access to essential equipment is not obstructed; and

(3) General hazards introduced into work area are made safe or corrected;

d. Cancelling a LOA IAW Annex SOP-S13E by:

(1) Verifying the reason for cancelling a LOA in part 3 of the form;

(2) Briefing the AE for the reason for cancelling the LOA; and

e. Closing a LOA IAW Annex SOP-S13E by:

(1) Visually inspecting the work area is clean and clear;

(2) Visually verifying the HV aware personnel has completed part 3 of the LOA form; and

(3) Completing part 4 of the LOA form to include:

(a) Signing both copies of part 4 of the PTW form; and

(b) Placing one completed PTW copy into the switchboard register;

(4) Completing the serial tracking list IAW Annex F SOP-S13 to include:

(a) Date completed; and

(b) Initials.

4. **Time Required:**

   a. EO – 2 x 50 min periods.

5. **Instructional Methods/Media:**

   a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

   a. Workstation;

   b. Workplace;

   c. References;

   d. DLN 3.0 access; and

   e. Computer with internet access.

7. **Test Details:**

   a. EO 001.06 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4, A5, A6, C1, C2

9. **Remarks:**

   a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Performing a Limitation of Access (LOA) procedure as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Preparing for a LOA to include: 1. Completing serial tracking list IAW Annex SOP-S13F; 2. Completing part 1 LOA form; and 3. Documenting special instructions and safety	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Limitation of Access Form

measures in part 1 LOA form if required;			
Issuing a LOA IAW Annex SOP-S13E by: a) Visually verifying PiC's credentials; b) Conducting a worksite walkthrough with PiC at POW to include: a. Extent of work to be carried out; b. Live HV equipment in or adjacent to the area where work is to be conducted; c. Safety brief at POW; d. Cordoning off the work area with caution tape; e. Obtaining the PiC's signature part 1 of the LOA form for acceptance of special instructions and safety measures if required; f. Obtaining the PiC's signature on part 2 of the LOA form for acceptance of scope and limitations of work; g. Making a duplicate copy of the LOA form under the same serial number to include: i. Placing the original copy in the switchboard	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2 Limitation of Access Form

register; and ii. Providing a duplicate copy to the PiC;			
Inspecting the worksite at the end of each working period or day to include: a) Flammable or hazardous materials are removed; b) Access to essential equipment is not obstructed; c) General hazards introduced into work area are made safe or corrected;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2 Limitation of Access Form
Cancelling a LOA IAW Annex SOP-S13E by: a) Verifying the reason for cancelling a LOA in part 3 of the form; b) Briefing the AE about the reason for cancelling the LOA;	Direct instruction	Online delivery via DLN 3.0	A1, A2, Limitation of Access Form
Closing a LOA IAW Annex SOP-S13E by: a) Visually inspecting the work area is clean and clear; b) Visually verifying the HV aware personnel has completed part 3 of the LOA form; and c) Completing part 4 of the LOA form to include: a. Signing both copies of part 4 of the PTW form; and b. Placing one completed PTW copy into the	Direct instruction	Online delivery via DLN 3.0	A1, A2, Limitation of Access Form

switchboard register; d) Completing the serial tracking list IAW Annex F SOP-S13 to include: a. Date completed; and b. Initials.			

- EO 001.06 shall be assessed in PC 001

EO 001.08 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Performing an HV restoration procedure as an APC IAW the pre-approved HVSS.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. Environment:

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform an HV restoration procedure as an APC IAW the pre-approved HVSS to include:

   a. Performing an HV restoration procedure as an APC IAW the pre-approved HVSS to include:

      (1) Initialling each restoration step in the HVSS form upon completion;

      (2) Removing any grounds on HV system;

      (3) Racking in the breaker;

      (4) Switching breaker to remote control;

      (5) Updating the HV stateboards; and

      (6) Verbally informing the AE and Duty Engineer that the HV system and equipment are now available.

4. **Time Required:**

   a. EO – 3 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.06 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A4

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Performing an HV restoration procedure as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Initialling each restoration step in the HVSS form upon completion	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form Annex A-Annex Z
Removing any grounds on HV system	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form Annex A-Annex Z
Racking in the breaker	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, High Voltage Switching Schedule Form

			Annex A- Annex Z
Switching breaker to remote control	Direct instruction	Online delivery via DLN 3.0	A1, A2
Updating the HV stateboards	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4
Verbally informing the AE and Duty Engineer that the HV system and equipment are now available	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4

- EO 001.07 shall be assessed in PC 001

EO 001.09 – Perform High Voltage Safe System of Work Procedures as an Authorized Person

1. **Performance:** Conducting APC handover certificate process as an APC.

2. **Conditions:**

    a. **Given:**

        (1) Computer;

        (2) DLN 3.0 access; and

        (3) References;

    b. **Denied:** Nil;

    c. **Environment:**

        (1) Any location; and

        (2) Day or night.

3. **Standard:** The trainee shall conduct an APC handover certificate process as an APC IAW SOP-S13H by:

    a. Informing state of HV on form;

    b. Noting state of HV equipment;

    c. Signing APC form; and

    d. Performing APC duties during handover.

4. **Time Required:**

    a. EO – 1 x 50 min periods.

5. **Instructional Methods/Media:**

    a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

    a. Workstation;

    b. Workplace;

    c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.08 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A4

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Conducting APC handover certificate process as an APC			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Informing state of HV on form	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, APC Handover Form
Noting state of HV equipment	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, APC Handover Form
Signing APC form	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, APC Handover Form
Performing APC duties during handover	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, APC Handover Form

- EO 001.08 shall be assessed in PC 001

PC 001 – Authorized Person Assessment

Assessment Activity
Strategy: Direct (via OJPR)
Media: Printed material and real equipment
Method: Practical assessment
Total Time: 145 mins per trainee 5 mins feedback per trainee
Environment: On ship, alongside, during normal working hours and not during extreme weather conditions
References: • NAVORD 11100-1 • HDW Class SEMS - SOP S13 – HV Switching Program • CSA Z462 Workplace Electrical Safety • DAOD 2007-1 - General Safety Program • Canada Labor Code Pt. 2 • General Safety Program – General Safety Standards, Volume 2 Chapter 17 - Electrical • HDW Class SSO AL11 25 April 20 • Standard for Electrical Safety in the Workplace 2018.
Pass/Fail Criteria: 80% and trainees must achieve SAT on all Critical checklist items (marked with an asterisk [*]). If a trainee fails to meet a critical checklist item, the assessment will cease. The impact of the failure is noted below.
Impact: Remedial training/re-test upon first unsuccessful PC. A TRB shall be held upon a second unsuccessful PC.
Limitations / Remarks: 1. Ship HV state: HV Dead

Assessment Details		
Input: Upon completion of stage 4 of training, the assessor will schedule the trainees' PCs. Assessor to provide assessment checklist to trainee before the assessment begins to allow the trainee to prepare for the assessment Before the assessment begins, the assessor will highlight that a failure to achieve SAT in any critical safety items in the evaluation will result in an automatic re-test. In addition, the assessor will provide a written copy of the PC scenario to thestudent. Once the trainee has read the scenario, the assessor will ask the trainee if they have any questions. Finally, the assessor will highlight they will act as the AP during the assessment. The trainee shall begin the assessment in the Engineering Office.		
Assessment Process: 1. Trainee completes HVSP preparation tasks IAW PC 001 Checklist 2. Trainee performs HV DG1 equipment isolation procedure IAW PC 002 Checklist 3. Trainee issues PTW to PiC CP IAW PC 001 Checklist 4. Trainee issues SFT to a PiC AP IAW PC 001 Checklist 5. Trainee closes PTW IAW PC 001 Checklist 6. Trainee performs HV DG1 restoration procedure IAW PC 001 Checklist 7. Trainee closes SFT IAW PC 001 Checklist		
Output: Upon completion of the assessment, the trainee will have demonstrated proficiency in independently performing the job of an AP		
Feedback / Debrief: Individual trainee feedback provided after the assessment.		
Assessment Resources		
OJPR coach Resources: • PC 001 Checklist • PC Scenario • Assessment Instructions • PP&S • Timer	Learner Resources: • PC 001 Checklist • PC Scenario • Arc flash 2 PPE • Pre-approved HVSS form • PTW form	To be Developed:

	• SFT form • PP&S • HDW Class SEMS SOP S-13 • Computer	
Continuous Improvement		
After the lesson is delivered, the instructor will use this section to add improvements, suggestions and comments.		

PC 001 Assessment Activity Checklist

Assessment Activity Checklist PC 001				
PC 001 Scenario: You are the APC on an AOPV. You are currently alongside conducting a basin trial with all four MDGs on load. You receive 48 hours advance notice of corrective maintenance. The contractor will be conducting maintenance, including an insulation test, on Diesel Generator No. 1.				
Candidate Name	**Assessor Name**	**Date**		**Course Session**
Item		**SAT**	**UN SAT**	**Comments**
1. Prep for HV Switching Program (30 mins)				
Obtain verbal approval from AE for HVSP				
***Conduct verbal briefing to duty staff** - Snr. Watch-keeper and Duty Engineer while alongside - Work and/or test being performed - Equipment/systems affected - Equipment restrictions				
Assessor to ask trainee whether an informative pipe is necessary in this scenario.				
Complete pre-approved HVSS form - Print form from common drive - Initiate serial tracking list IAW Annex SOP-S13F • Date issued • Serial number • Form type • Purpose of system/equipment • PIC • Key Safe - Make duplicate copy of HVSS under same serial number				

- Second AP must review and sign the HVSS form Part 1 under “Checked by” (only checking the procedure on the paper)			
Complete PTW Form - Print PTW form from common drive - Complete Part 1 PTW form			
Complete SFT Form - Print SFT form from common drive - Complete Part 1 SFT form			
2. Perform HV Equipment Isolation (60 mins)			
- Obtain AE initials beside step 1 of pre-approved HVSS - Obtain AE signature for PTW - Obtain AE signature for SFT			
*Open HV VCB MV2DG1 remotely via IPMS and record decay time (5 mins)			
*Muster tools, padlocks, keys, signs, barrier tape, appropriate arc flash and arc flash PPE			
***Setup a safe working area with restricted access in Port Motor Room** - Roping off area - Affixing signage - Set up red DANGER tape, limits of approach			
***Visually inspect cat 2 arc flash PPE** - Arc-rated coveralls - Arc-rated balaclava - Arc-rated face shield - Heavy-duty leather gauntlets - Rubber gloves - HV arc-flash rated leather boots			

*From inspection window, confirm breaker indicator is OPEN *HDW-182 MV2 labels*			
*Switch MV2DG1 CB control selector to LOCAL MANUAL control *on switchboard, HDW-183 (black selector switch, no key)*			
*Remove CB Control Selector Switch key			
*Rack Out Breaker MV2DG1 *HDW-182, 3 fortress keys, open window, rotate counter-clockwise with black handle until green light comes on when breaker is fully racked out*			
*Apply caution sign and padlock to racking mechanism and record Lock No.			
*Remove Generator Panel Key B6-1 (breaker racking-in interlock) and place in the key exchange box			
*Insert Generator Panel Key B6-3 (CME closing interlock) *insert key in centre, rotate to the right*			
*Wait 5min minutes for decay time (see step 2 for time) to elapse before applying Circuit Main Earth			
*Apply CME and record time CME applied *use wrench to pull out CME from void, apply lock (green lock) HDW-182 (far right)*			
*Apply caution sign to CME mechanism access port *centre port (Caution tag)*			
*Remove Generator Panel Key B6-2 (CME opening interlock)to be placed in the key exchange box *remove key from far right (turn left)*			
*Confirm CME is in the CLOSED position (CLOSED/RED) position			

indicator on CME mechanism access port *red bar inside CME access port is red (if not applied, green)*			
*Ensure area is clean and secure with proper signage and safety barriers rigged -Obtain CP qualified member above to act as safety member • Brief member • Provide safety cane			
*Communicate with the Lockout/Tagout (LOTO) coordinator whenever there is a requirement for a LOTO, such as low voltage, mechanical, fluid or gas isolation. • Include LOTO form and lock box number on HVSS form			
*Loosen bolts on generator connection box • Leave some bolts in place			
***Don cat 2 PPE** - Arc-rated coveralls - Arc-rated balaclava - Arc-rated face shield - Heavy duty leather gloves - Arc-rated rubber gloves - Arc-rated leather footwear			
Take off bolts on generator connection box • Ensure bolts are stored in a safe location			
***Prove HV circuit dead** -Test Potential Indicator (PI) -Testing connections with PI to include: • Phase to phase • Phase to ground			

- Visually check all connections in connection box for high voltage - Re-test PI			
***Update HV stateboards** - Shipwide HV stateboard - Equipment HV stateboard			
3. Issue PTW to PIC CP (5 mins)			
Visually verify PIC - HV work credentials; - Current standard first aid - Personal key safe padlocks; and - Padlock identification tags			
Apply Additional Earth (Spider) as required by PiC. **Additional Earth Applied**: Yes / No			
Muster all HVSS keys and place in PTW Key-Safe			
Lock multi-lock device on SWBD key safe to be locked using APC padlock. • Ensure PIC places his/her personal padlock on multi-lock device. • Retain APC padlock key.			
Obtain PIC signature under HVSS isolation sketch			
***Conduct HV system walkthrough at POW** - Isolation points - Earthing points - HVSP devices used - Safety arrangements at POW			
-Obtain PIC's initial part 1 PTW form			
***Physically prove HV circuit is DEAD at POW to the PIC** - Place bare hands on all connections			

- Ask PIC if there is anything else to prove dead			
Issue PTW - Obtain PIC's signature part 2 PTW form			
Store PTW in Switchboard Register; Give copy to PIC			
Assessor to inform trainee that the external contractor has completed the work outlined in the PTW.			
4. Close PTW (5 mins)			
Visually inspect work area and equipment: - Clean work area - Account for and remove all tools/materials accounted for and removed - Ensure all equipment covers are replaced and secure			
Visually verify PIC has completed Part 3 PTW form			
Complete Part 4 of PTW form			
Ensure that PIC removes padlock from key safe			
Assessor to inform trainee that the external contractor is ready for Insulation Testing			
5. Issue SFT to a PIC AP (5 mins)			
Visually verify PIC to include: - HV work credentials; - Personal key safe padlocks; and - Padlock identification tags - Appropriate PPE (CAT 2 for DG1)			
*Brief PIC AP at POW - Purpose of test - HV equipment made safe			

- Isolation points; - Earthing points; - HVSP devices used - Safety arrangements at POW, including limits of approach and PPE			
Update HV Stateboards; Equipment HV stateboard for DG1			
- Obtain PIC's initials part 1 SFT form; - Obtain PIC's signature part 2 SFT form			
Assessor will inform trainee that insulation testing is complete.			
6. Close SFT (10 mins)			
- Visually inspect work area for cleanliness - All tools/materials removed			
Visually verify PIC has completed Part 3 SFT form			
Complete Part 4 of SFT form			
Ensure PIC removes padlock from the key safe			
7. Perform Restoration Procedure (30 mins)			
*Insert generator panel key B6-2 (CME opening interlock)			HVSS 19
*Remove CME caution sign (tag)			HVSS 20
*Remove applied CME (disengage)			HVSS 20
*Remove generator panel key B6-3 (CME closing interlock)			HVSS 20
*Insert generator panel key B6-1 (breaker racking in interlock)			HVSS 21
*Remove padlock and caution sign from breaker racking mechanism access port			HVSS 22
*Rack in breaker MV2DG1			HVSS 22
*Insert CB control selector switch key			HVSS 23

*Switch MV2DG1 to remote control			HVSS 23
Take down/Muster tools, padlocks, keys, PPE, signs, barrier tape, etc.			HVSS 24
Ensure non-HV Lockout/Tagout is completed.			HVSS 25
□ Ensure all documents are removed from display points. □ Ensure completed PTW, SFT and HVSS are retained in Switchboard Register. Complete serial tracking list IAW Annex F SOP-S13 • Date completed • initials			HVSS 26
*Update HV stateboards -Equipment HV stateboard			
Inform the AE that work is complete and DIESEL GENERATOR No.1 breaker is now available.			HVSS 27
Additional Comments			
Strengths:		Areas of development:	
The candidate has received a debrief and seen the content of this assessment			
Overall assessment (circle one)		PASS	FAIL
	Name	**Date**	**Signature**
Assessor			
Candidate			

PC 001 – Authorized Person Training Plan

Training Activity
Strategy: OJPR coach, real equipment, printed material
Media: OJPR coach, real equipment, printed material
Method: 1. HV Training Onboarding Meeting 2. eLearning Course 3. Virtual Task Trainer Training Activity 4. Physical Tour of HV Spaces on ship 5. OJPR Signature: Demonstration & Practice Activity
Total Time: 1215 minutes (4 days)
Environment: Distributed Learning, Virtual Task Trainer, and HMC Ship
References: • NAVORD 11100-1 • HDW Class SEMS - SOP S13 – HV Switching Program • CSA Z462 Workplace Electrical Safety • DAOD 2007-1 - General Safety Program • Canada Labor Code Pt. 2 • General Safety Program – General Safety Standards, Volume 2 Chapter 17 – Electrical • HDW Class SSO AL11 25 April 20 • Standard for Electrical Safety in the Workplace 2018 •
Pass/Fail Criteria: N/A
Impact: Nil.

Limitations / Remarks:

1. Availability of HV SMEs and AOPV ship to conduct training that impacts scheduling
2. There are no ECs for this PO due to the infancy/limited information of the capability and lack of training resources/information
3. Ship HV state: HV Dead
4. The VTT will not be available as part of serial 1 and is expected to be available starting serial 2
5. AP eLearning lessons have not been validated by HQ

Training Details

Input:

Trainee gets posted to an AOPV as a member of the Naval Technical Department Team and reports to SEPM

Activity Process:

Stage 1 – HV Training Onboarding Mtg (30 mins)

- Trainees conduct a one-on-one mtg with SEPM about their HV individual training and education roadmap using HV training requirement resources and student guide

Stage 2 – Complete eLearning Course (unlimited)

- Trainees complete the AP eLearning course from home/workplace using their own personal digital device or work-issued tablet. The following eight DL lessons are to be completed in the following sequence:
    - L1: General introduction to HV
    - L2: Electrical Hazards & Arc Flash
    - L3: HV systems and compartments
    - L4: HV states, stateboards, and pipes
    - L5: HVSP locking devices
    - L6: HV testing apparatus and types of testing
    - L7: HV SSOW administration and system control
    - L8: HV switching program procedures

Stage 3 – Virtual Task Trainer – DG1 Isolation Procedure & EC 001 (120 mins)

Trainees complete Diesel Generator 1 (DG1) training activity via VTT. The VTT will adopt a gated approach; trainees can only move onto the next stage of VTT training activity if they meet the minimum standards for each gate. This

activity will enable trainees to build skill and confidence in isolating DG1 virtually before being assessed on ship as part of OJPR.

***** NOTE – There is no VTT currently available during the creation of this OJPR checklist (June 2022). State 3 shall refer to instructions when a VTT becomes available for HV training.***

Stage 4 – HV Physical Tour (60 mins)

- Trainees receive a physical tour by OJPR coach IAW with tour guide resource

Stage 5 – On Job Training

<u>Visually Inspecting Cat 2 & 4 PPE (60 mins)</u>

Demonstrate

- OJPR coach initiates training activity by emphasizing key safety points, including it is the duty of each personnel to visually inspect their issued arc flash PPE before the commencement of any HV work, personnel are never to commence HV work with damaged and/ or unclean arc flash PPE, personnel must report any arc flash PPE damages to their supervisor immediately upon discovery to initiate the replacement process, and the APC must always visually inspect the assigned PiC AP's arc flash PPE before he/she may commence HV work. OJPR coach then conducts an explanation and demonstration at the same time of how to visually inspect PPE.

Practice Activity

- Trainees practice visually inspecting cat 2 & 4 PPE with a partner while under OJPR coach supervision. Partners are to provide feedback during the activity, and the OJPR coach is to provide ongoing feedback to individuals/groups during the activity
- Once the OJPR coach is confident trainees can visually inspect cat 2 & 4 PPE independently, the group will move on to the next step.

OJPR Signature 1

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in visually inspecting PPE.

Donning Arc Flash Cat 2 & 4 PPE (60 mins)

Demonstrate

- OJPR coach initiates the training activity by emphasizing key safety points, including AP qualified members who may work on dead and live HV equipment. Personnel shall never commence HV work without the min required PPE. OJPR coach also communicates where to find arc flash 4 PPE on ship. OJPR coach then explains and demonstrates at the same time how to don arc flash cat 2 & 4 PPE.

Practice Activity

- Trainees practice donning cat 2 & 4 PPE with a partner while under OJPR coach supervision. The partner provides feedback during the activity, and the OJPR coach provides ongoing feedback to individuals/groups during the activity.
- Once the OJPR coach is confident trainees can independently don cat 2 & 4 PPE gear, the group will move on to the next step.

OJPR Signature 2

- The OJPR coach signs the trainees' OJPR package if they have successfully demonstrated proficiency in donning arc flash PPE.

Supervising a PTW as a PiC AP (60 mins)

Demonstrate

- OJPR coach initiates the activity by providing each trainee with an actual PTW form that has been completed in the past. The OJPR coach then provides an overview of the PTW procedure and examples of common PTW scenarios. The OJPR coach then demonstrates while explaining how to supervise a PTW as a PiC AP.

Practice Activity

- Trainees practice completing the PTW procedure with a partner, and during the activity, one partner acts as the APC and evaluates their partner using the checklist while under the supervision of the OJPR coach. The OJPR coach provides ongoing feedback to individuals/groups during the activity. Once the OJPR coach is confident trainees can independently perform the PTW procedure, the group will move on to the next step.

OJPR Signature 3

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in supervising a PTW as a PiC AP

Supervising a SFT as a PiC AP (60 mins)

Demonstrate

- OJPR coach initiates the activity by providing each trainee a copy of a real SFT form that has been completed in the past. The OJPR coach then provides an overview of the SFT procedure and examples of common SFT scenarios. The OJPR coach then demonstrates while explaining how to supervise a SFT as a PiC AP.

Practice Activity

- Trainees practice completing the SFT procedure with a partner, and during the activity, one partner acts as the APC and evaluates their partner using the checklist while under the supervision of the OJPR coach. The OJPR coach provides ongoing feedback to individuals/groups during the activity. Once the OJPR coach is confident trainees can independently perform the SFT procedure, the group will move on to the next step.

OJPR Signature 4

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in supervising a SFT as a PiC AP

Preparing for HV Switching Program as an APC (60 mins)

Demonstrate

- The OJPR coach initiates this activity by providing a copy of a HVSP prep checklist. The OJPR coach demonstrates where to access all the HVSP forms on the computer. The OJPR coach then provides an overview of the HVSP preparation and various scenarios. The OJPR coach demonstrates while explaining how to prepare for the HV switching program as an APC IAW pre-approved DG1 HVSS.

Practice Activity

- Trainees will practice completing the HVSP prep for DG1 HVSS with a partner while the OJPR coach provides ongoing feedback to individuals/groups during the activity
- Once the OJPR coach is confident trainees can independently perform the HVSP prep, the group will move on to the next step.

OJPR Signature 5

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in preparing for the HV switching program as an APC

Performing HV Equipment Isolation Procedure as an APC (90 mins)

Demonstrate

- The OJPR coach initiates the activity by providing an overview of the isolation procedure, isolation scenarios and key safety points. The OJPR coach demonstrates while explaining how to isolate the equipment IAW pre-approved DG1 HVSS.

Practice Activity

- OJPR coach splits members into groups of two, and each will perform the DG1 HVSS isolation procedure separately under the supervision of the OJPR coach.
- Once the OJPR coach is confident trainees can independently perform the isolation procedure, the group will move on to the next step.

OJPR Signature 6

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing HV equipment isolation procedures as an APC.

Performing HV Restoration Procedure as an APC – DP (60mins)

Demonstrate

- OJPR coach initiates activity by providing an overview of the restoration procedure and various restoration scenarios. The OJPR coach demonstrates while explaining how to restore the HV equipment IAW DG1 HVSS form.

Practice Activity

- OJPR coach splits members into groups of two, and each group will perform the DG1 HVSS restoration procedure separately under the OJPR coach supervision
- Once the OJPR coach is confident trainees can independently perform the isolation procedure, the group will move on to the next step.

OJPR Signature 7

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing HV restoration procedures as an APC.

Performing PTW Procedure as an APC (60 mins)

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed PTW example IAW PC 002 scenario. The OJPR coach demonstrates while explaining how to perform the PTW procedure IAW PC 002 checklist.

Practice Activity

- OJPR coach provides trainees with a blank PTW form and trainees complete the PTW procedure IAW PC 002 checklist individually while under OJPR coach supervision. Trainees then submit the completed form to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee.
- Once the OJPR coach is confident trainees can independently perform the PTW procedure, the group will move onto the next step.

OJPR Signature 8

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the PTW procedure as an APC.

Performing SFT Procedure as an APC (60 mins)

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed SFT example IAW PC 002 scenario. The OJPR coach demonstrates while explaining how to perform the SFT procedure IAW PC 002 checklist.

Practice Activity

- OJPR coach provides trainees with a blank SFT form, and trainees complete the SFT procedure IAW PC 002 checklist individually while under OJPR coach supervision. Trainees then submit the completed form to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee.

- Once the OJPR coach is confident trainees can independently perform the SFT procedure, the group will move on to the next step.

OJPR Signature 9

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC.

<u>Performing LOA Procedure as an APC (60 mins)</u>

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed LOA example. The OJPR coach demonstrates while explaining how to perform a LOA procedure.

Practice Activity

- OJPR coach provides trainees with a blank LOA form and new LOA scenario. Trainees complete the LOA procedure individually while under OJPR coach supervision. Trainees then complete the form and submit to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee
- Once the OJPR coach is confident trainees can independently perform the LOA procedure, the group will move on to the next step.

OJPR Signature 10

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC.

<u>Performing APC Handover Process – DP (45 mins)</u>

Demonstrate

- OJPR coach initiates this activity by providing an overview of the APC handover certificate process. The OJPR coach leads the trainees to the main switchboard location to demonstrate while explaining how to visually inspect and record the I/O, mode and condition of all breakers.

Practice Activity

- OJPR coach provides a blank APC handover certificate form to all trainees. The trainees practice completing the APC handover certificate while under the supervision of the OJPR coach. Trainees complete the form and submit

to the OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee.

- Once the OJPR coach is confident all trainees can complete the APC handover process form independently, the group will move on to the next step.

OJPR Signature 11

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC.

Output:

At the completion of this training plan, the trainee will have demonstrated proficiencies to perform HV tasks as an Authorized Person and proceed to the PC 002 summative assessment.

Feedback / Debrief:

- Trainees will have a training completion brief meeting with the SEPM to include:
  - Training debrief
  - Verbal confirmation trainee will be awarded associated MITE and NQUAL for completing HV Safe System of Work course
  - Completion of certificate of appointment authorized person by AE and APC endorsed SEPM

Training Resources		
OJPR coach Resources: • PC 001 • PPE • HV Tools • Physical Tour Guide • HVSP forms • Annex SOP S13K – Certificate of Appt Process AP • Computer • Printer • Training schedule • HV training student guide	Learner Resources: • PC 001 • PPE • HV Tools • PTW form • PTW scenario • SFT form • SFT scenario • LOA form • LOA scenario • PTW checklist • LOA checklist	To be Developed: • LOA scenario

	• Personal Digital Device or work computer • Training schedule • HV training student guide	
Continuous Improvement		
After the lesson is delivered, the instructor will use this section to add improvements, suggestions and comments.		

ANNEX A – MAIN REFERENCES

A = CAF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other

Ref Code	Publication
A1	NAVORD 11100-1
A2	HDW Class SEMS - SOP S13 – HV Switching Program
A3	DAOD 2007-1 - General Safety Program
A4	HDW Class SSO A11
A5	Canada Labour Code Pt. 2
A6	General Safety Standards, Volume 2 Chapter 17 - Electrical
A7	Damage Control Instructions 03/20 Home Port Alongside
C1	CSA Z462 Workplace Electrical Safety 2021
C2	Standard for Electrical Safety in the Workplace 2018

ANNEX B – TASK LIST

Task No	Task Statement	Train / No Train	NT Reasoning	PO 001 - AP
A1	Supervise PTW	T		AP
A2	Supervise SFT	T		AP
A3	Conduct HVSP	T		AP
A4	Perform HVSS equipment isolation	T		AP
A5	Perform HVSS restoration	T		AP
A6	Conduct APC/SEPM handover process	T		AP
A8	Issue PTW	T		AP
A9	Close PTW	T		AP
A10	Cancel PTW	T		AP
A11	Issue SFT	T		AP
A12	Close SFT	T		AP
A13	Cancel SFT	T		AP
A14	Issue LOA	T		AP
A15	Close LOA	T		AP
A16	Cancel LOA	T		AP

A17	Conduct HV safety program development and evaluation technique profile	NT	Task rarely performed	AP
A18	Develop HV equipment sketches for HVSS	NT	Take rarely performed	AP
A19	Manage HV equipment, devices, and accessories	NT	Prior learning	AP

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 11 Mar 2021

Location: Virtual via MS Teams

Board Members:

Position	Rank and name	Role / Unit	Dates
Chair	PO1 Aaron Reid	Sr. Electrical Propulsion Mgr on HMCS HDW	11 Mar 21
SME	PO1 Marvin Murphy	LOTO Coordinator, FMF Cape Scott	11 Mar 21
SME	PO2 Michael Lamanque	Sr. Mech Specialist on HMCS HDW	11 Mar 21
TDO	SLt Sebastian Seo	TDO, Training Dev Centre Pacific	11 Mar 21
Secretary	Ms. Marilyn Carey	Civilian	11 Mar 21

Board Objective: To finalize the HV Safe System of Work QSP

Decision/Event	Discussion / Detail
Day 1: March 11, 2021	
SMEs	Given high voltage is so new to the RCN, there are no real SMEs for HV in the RCN
No task list	No approved HV task list from AOPV PMO or Occupational Authority or updated MAR Tech JTAR (last updated in 2019). Board referenced SEMS SOP-S13 document to identify HV tasks and based on reality of the job.
Exclude devising	PO 2 Lamanque commented that there may be some circumstances where an APC may need to devise a hand written sketch outside of the sketches provided on the pre-authorized

Decision/Event	Discussion / Detail
sketches for HVSS in QSP	HVSS form. Discussion about need to include this in QSP. HVSS pre-approved forms to be verified through QA on Mary Brooke (Seo). Decision: No train due to infrequency of event, prior training
Exclude HV maintenance tasks on QSP.	Discussed whether performing non-HV maintenance tasks should be inserted into QSP PO 002 (AP). Reid: methods for maintenance not changing depending on HV equipment. Only change is locking out. Lamanque: Maintenance is done HV Dead. Seo: MarTechs have prior training on maintenance. Murphy: I concur.
	Lamanque: Assume HV Live state until proven dead. Should this be included? Reid: Yes, I agree
Exclude performing SFT insulation testing	For insulation testing, prior learning for MAR Tech elecs on smaller circuits – already know how to do this (Reid). In reality, APs have been supervising contractors – this is also what the contract stipulates with external contractors (Murphy). Exclude from QSP
Exclude performing maintenance during SFT.	PO 002 (6) Discussion – SFT while at Sea. Lamanque: May be a situation at sea where ship staff will conduct work under an SFT. Reid: Yes, SFT includes running the machinery for testing purposes, e.g. propulsion converters updates on recent sail. Seo: Did you get formal training through Calhoon? Reid: Everything I've learned has been on my own. That's what we do as electricians.
Remove "HV safe" from PO 002 d.(6)	Reid: Only APC is to make the informative pipe for "HV Safe" during a damage control event, but the caveat is Duty Engineer would need to make this pipe while alongside and the ship is HV Live. Murphy: This is only related to when the ship is HV Live.
Change "switchboard safe" to "switchboard register" PO 002 d.(7)	Decision to change to "switchboard register" instead of "switchboard safe" because there is no place to mount the document safe on the wall on the AOPV ship because of fuel tanks, etc. (Reid)
PO 002 e. (10)	Discussion of role of safety member. Part of electrical training to never work on live circuits alone. If you need to test something, you would have safety member with 10 ft safety cane in case of accident (Murphy). Action of safety member = NO TRAIN (Seo). "Leave this one with me" (Seo)

Decision/Event	Discussion / Detail
PO 002 e. (8)	Placing all HVSS keys in switchboard key safe moved from (10) to (8). Important to put in QSP (Reid).
PO 002 f. (a.iv)	Added “visually inspecting worksite PPE.” Worksite PPE, not necessarily Arc Flash rated PPE needed. (Reid)
PO 002 f. (4.c.)	Discussion of signing off on both copies of PTW, including PiC’s copy. (Murphy) Both need to be completed. Added f. (4.c.) that one completed copy of PTW needs to go into switchboard register.
PO 002 g. (2)	AP is always the PiC but they can either undertake or supervise – focus on supervising (Seo). Our standard for this QSP is supervising, not doing. (Seo) Confirmed by Reid.
PO 002 h. 2(b) Are you always cordoning off the area with caution tape? (Murphy)	Not all the time, spaces are so big – never did an LOA in MCR, all LOAs so far main machinery room. These spaces are massive, HV is in one small area. Only once cordon off (Reid). There should never be concurrent work going on (Murphy). Decision to train personnel to cordon off. (Seo) Confirmed by Reid.
PO 002 i.	Added “Update HV Stateboards” (Murphy)
PO 002 k.	Do we need to train ppl: Managing all HV equipment, devices and accessories as an APC to include: Visually inspecting all HVSP equipment, devices and accessories periodically IAW Equipment Inspection Guidelines to include: Cleanliness; Damage; and Security. Tracking HVSP equipment, devices and accessories accessed by personnel; and Ordering replacement HVSP equipment, devices and accessories. Decision: No. Already trained duty engineers. This is no different. (Reid, Lamanque) NO TRAIN – Prior learning
PO 002 l.	Conducting a SEPM handover IAW Annex SOP-S13G with the oncoming Senior Electrical Propulsion Manager No Train as SEPM handover is same as APC Handover. It would be AE level.

Decision/Event	Discussion / Detail
	Completing HV safety program dev and evaluation technique profile – task rarely performed?
PO 002 m.	Completing HV safety program development and evaluation technique profile form IAW Annex SOP-S13O Not sure if this needs to be trained. Personally, I will do it. (Reid). Seems to be an experiential learning piece (Lamanque) NO TRAIN – task is done rarely. No SME experience with this. (Seo)
PO 001-3a.	Remove "Visually inspect and don PPE" from CP training. I've never seen a situation where the CP needs to don the PPE (Reid).
PO 001-3	Remove printing HV forms from CP level. AP would print these forms.
PO 001-3	"Completing" HVSS changed to "Reviewing" HVSS (Murphy).
PO 001-3	Conducting HV work on dead HV equipment to include: Changing air filter; Changing technical water; Changing oil; and Changing light bulb No train since MarTech Mechs already have prior training on CM and PM. (Reid, Lamanque, Murphy)
PO 001-3	Remove "Supervising work performed by HV aware personnel to include: a. Work being completed IAW LOA; Safety measures being followed IAW HV SSOW; and Work area is clean upon completion of work.
PO 003-3a,b	Train (a) based on SME response. AE must know the process. (Reid, Murphy) (b) These are different processes – all separate certificates, handwritten (Reid). CP Certification: WEng CANNOT work on any HV equipment; Mec's can only work on DEAD HV equipment; AP's can work on LIVE HV equipment– long process, must train (Reid).
PO 003-3c,d	Authorizing all High Voltage Switching Schedule (HVSS) work by: Visually checking the HVSS form is complete; Initialling step 1 of HVSS form. Authorizing all HVSP work by: Visually checking HVSP forms are complete;

Decision/Event	Discussion / Detail
	Signing HVSP forms to include: (a) PTW; SFT; and HVSS No Train (c) – leaning on APC, AE needs to know what work is being done and rationale behind it (Reid). 90% of work, they know it's coming (Murphy). Already trained in AP module (Seo). No Train (d) – Already trained in AP course. (Seo) Confirmed by Reid, Lamanque and Murphy.
PO 003-3e,f	Visually inspecting HVSP PPE and testing equipment maintenance records periodically; Conducting monthly review of switchboard register to include: (1) Visually inspecting volumes 1 and 2; Signing in volume 1 and 2 binders; and Forwarding volumes 1 and 2 to CO for quarterly review and signature. No Train (e) – just making sure records are being kept (Reid, Lamanque) Records not signed off – nothing to train, but need to be aware (Murphy). These should fall under "Responsibilities" for AE training (Lamanque, Seo) No Train (f) – easy task (Seo), knowledge base (Reid). Navord indicates CO signs monthly. (Murphy)
PO 003-3g, h	Conducting periodic on-site safety audits by: (1) Visiting an HV work site; Visually inspecting safety measures are being followed. Responding to HV near miss incidents to include: (1) Completing HV Near Miss Report SOP-S13P; Handing-over report to Unit General Safety Officer; Placing copy in switchboard register binder. NO Train (g) – prior learning in AP (Lamanque) NO Train (h) – admin (Lamanque), UGSO generates report (Murphy) simple task (Seo) AE Half day course, knowledge based on Sultan with test at the end (Reid)
PO 003-3 i,j	Obtaining permission from CO for the emergency removal of HVSP locking devices. Authorizing SEPM and APC handover certificate to include:

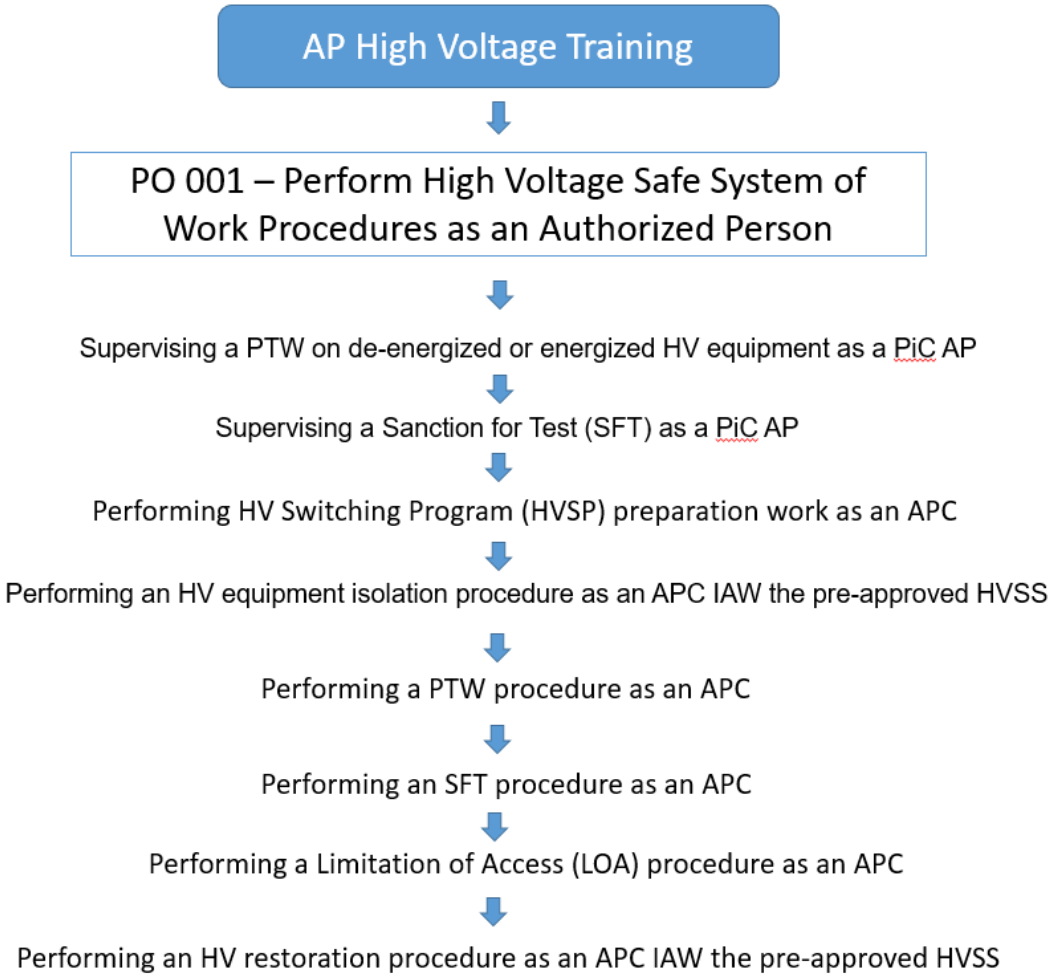
Decision/Event	Discussion / Detail
	a. Visually checking handover certificate forms to include: Senior Electrical Propulsion Manager handover certificate SOP-S13G APC handover certificate SOP-S13H Signing part 3 certificate of transfer No Train (i) simple task and/or not done rarely. Covered in LOTO system (Reid) No Train (j) SME's agreed.
PO 003-3 k,l	Approving the master key register by: a. Visually checking SOP-S13M form is complete; and Initialling the SOP-S13M form. Verbally briefing the captain on ship of deviations to approved safety rules. No train (k) – simple task, rarely done. Keys will never change b/c we use same keys and locks all the time. Takes awhile to set up. HDW every key and lock has one place to go. No train (l) – Already have training on how to talk to CO (Reid, Lamanque, Murphy)
AE Assessment?	Test should ask for references and where to find them (Murphy). Reid agreed. Scenarios? No need to request HMS Sultan Test – not relevant to AOPV (Reid)
Knowledge of references?	SEMS S-13, CSA bits and pieces, Canada Labour Code bits and pieces, Discuss later to figure out which pieces.
Day 2: March 17, 2021	
Assessment of PO 002	Regarding assessment, should the AP trainee be assessed on: (a) When closing the PTW, visually inspect work area, including clean work area, all tools and materials removed? (b) When closing the SFT, visually inspect work area, including clean work area, all tools and materials removed? (c) When closing the SFT, inform the AE and Duty Tech that work is complete, and DG 1 breaker is now available? Yes to a, b, c. These are all important duties for the APC. (Murphy, phone call with Marilyn). After the completion of a PTW or SFT, the PiC is responsible to ensure the worksite is

Decision/Event	Discussion / Detail
	cleaned up and the APC will inspect the worksite to ensure that it is complete before all final signatures. (Lamanque, email)

ANNEX D – QSPWB SCALAR

The mapping of the task analysis from the QSPWB is detailed in the standard paragraphs of the POs in Chapter 3 of this QSP, therefore, no scalar is inputted here.

The mapping of the instructional analysis from the QSPWB is detailed in the standard paragraphs of the EOs in Chapter 3 of this QSP, therefore, no scalar is inputted here.



ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

On-the-Job Training (OJT)
OJT is a component of formal training in which personnel learn tasks while performing their actual duties on a job. It is guided by performance objectives and checks.

The On-the-Job Performance Record (OJPR)
As part of HV interim training, the designated AOPV ship is responsible for conducting OJT, assessing learners and documenting whether or not the objectives were attained in the learner's OJPR. The designated AOPV ship reports the results of the OJT back to the training establishment responsible for the training programme. The training establishment enters the results on the individuals' record and awards the qualification upon successful completion.

Prior to commencing OJT, personnel are expected to have completed stages 1 to 4 below:

Stage 1 – HV Training Onboarding Mtg
Stage 2 – AP eLearning Course
Stage 3 – Virtual Task Trainer – DG1 Isolation Procedure (starting Spring 22 trg serial)
Stage 4 – HV Physical Tour on ship
Stage 5 – OJT on designated AOPV

(Certificate of Qualification)

Name and Initials	Rank	SN

Date Training Started	Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS
One copy of this OJPR, referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee to record and review individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

As the trainee completes each Performance Objective (PO) to the required standard, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues, they are directed to contact Standards at the appropriate TE for guidance.

If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under “Additional Duties” at the end of the OJPR.

PROGRESS REVIEW

Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers),1

OJPR Checklist - Authorized Person

ID	Practice Activities	OJPR Reviewed By (Rank, Position, Name)	Signature	Date
1	Visually inspecting arc flash PPE			
2	Donning arc flash PPE			
3	Supervising a PTW as a PiC AP			
4	Supervising a SFT as a PiC AP			
5	Preparing for HVSP as an APC			
6	Performing HV equipment isolation procedure as an APC			
7	Performing HV restoration procedure as an APC			
8	Performing PTW procedure as an APC			
9	Performing SFT procedure as an APC			
10	Performing LOA procedure as an APC			
11	Performing APC handover process			

TRAINING COMPLETION CERTIFICATE
Date training completed:

CERTIFICATION OF QUALIFICATION
I certify that _____ has completed training to the standards stated in the HV Safe System of Work QSP: PO 002 – Perform HV SSOW Procedures as an Authorized Person. The member is qualified to carry out the duties of an Authorized Person.

OJPR Coach Signature: _____ Date: _____

HOD Signature: _____ Date: _____

Training Activity Details

Stage 5 – OJT

<u>Activity 1: Visually Inspecting Cat 2 & 4 PPE</u>

Demonstrate

- OJPR coach initiates training activity, by emphasizing key safety points including it is the duty of each personnel to visually inspect their issued arc flash PPE before the commencement of any HV work, personnel are never to commence HV work with damaged and/ or unclean arc flash PPE, personnel must report any arc flash PPE damages to their supervisor immediately upon discovery to initiate the replacement process, and the APC must always visually inspect the assigned PiC AP's arc flash PPE before he/she may commence HV work. OJPR coach then conducts an explanation and demonstration at the same time of how to visually inspect PPE

Practice Activity

- Trainees practice visually inspecting cat 2 & 4 PPE with a partner while under OJPR coach supervision. Partners are to provide feedback during the activity and the OJPR coach to provide ongoing feedback to individuals/groups during the activity

- Once the OJPR coach is confident trainees can visually inspect cat 2 & 4 PPE independently, the group will move onto the next step

OJPR Signature 1

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in visually inspecting PPE.

<u>Activity 2: Donning Arc Flash Cat 2 & 4 PPE</u>

Demonstrate

- OJPR coach initiates the training activity by emphasizing key safety points including AP qualified members may work on dead and live HV equipment, personnel are never to commence HV work without the min. required PPE, and where to find the min required PPE sighting levels on HV equipment. OJPR coach also communicates where to find arc flash 4 PPE on ship. OJPR coach then explains and demonstrates at the same time how to don arc flash cat 2 & 4 PPE

Practice Activity

- Trainees practice donning cat 2 & 4 PPE with a partner while under OJPR coach supervision. The partner provides feedback during the activity and the OJPR coach provides ongoing feedback to individuals/groups during the activity

- Once the OJPR coach is confident trainees can independently don cat 2 & 4 PPE gear, the group will move onto the next step.

OJPR Signature 2

- The OJPR coach signs the trainees' OJPR package if they have successfully demonstrated proficiency in donning arc flash PPE

Activity 3: Supervising a PTW as a PiC AP

Demonstrate

- OJPR coach initiates the activity by providing each trainee a real PTW form that has been completed in the past. The OJPR coach then provides an overview of the PTW procedure and examples of common PTW scenarios. The OJPR coach then demonstrates while explaining how to supervise a PTW as a PiC AP.

Practice Activity

- Trainees practice completing the PTW procedure with a partner and during the activity one partner acts as the APC and evaluates their partner using the checklist while under the supervision of the OJPR coach. The OJPR coach provides ongoing feedback to individuals/groups during the activity. Once the OJPR coach is confident trainees can independently perform the PTW procedure, the group will move onto the next step.

OJPR Signature 3

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in supervising a PTW as a PiC AP

Activity 4: Supervising a SFT as a PiC AP

Demonstrate

- OJPR coach initiates the activity by providing each trainee a copy of a real SFT form that has been completed in the past. The OJPR coach then provides an overview of the SFT procedure and examples of common SFT scenarios. The OJPR coach then demonstrates while explaining how to supervise a SFT as a PiC AP.

Practice Activity

- Trainees practice completing the SFT procedure with a partner and during the activity, one partner acts as the APC and evaluates their partner using the checklist while under the supervision of the OJPR coach. The OJPR coach

provides ongoing feedback to individuals/groups during the activity. Once the OJPR coach is confident trainees can independently perform the SFT procedure, the group will move onto the next step.

OJPR Signature 4

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in supervising a SFT as a PiC AP

Activity 5: Preparing for HV Switching Program as an APC

Demonstrate

- The OJPR coach initiates this activity by providing a copy of the DG1 HVSS to all trainees. The OJPR coach demonstrates where to access all the HVSP forms on the computer. The OJPR coach then provides an overview of the HVSP preparation and various scenarios. The OJPR coach demonstrates while explaining how to prepare for the HV switching program as an APC IAW DG1 HVSS

Practice Activity

- Trainees will practice completing the HVSP prep for DG1 HVSS with a partner while the OJPR coach provides ongoing feedback to individuals/groups during the activity

- Once the OJPR coach is confident trainees can independently perform the HVSP prep, the group will move onto the next step.

OJPR Signature 5

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in preparing for the HV switching program as an APC

Activity 6: Performing HV Equipment Isolation Procedure as an APC

Demonstrate

- The OJPR coach initiates activity by providing an overview of the isolation procedure, isolation scenarios and key safety points. The OJPR coach demonstrates while explaining how to isolate the equipment IAW DG1 HVSS

Practice Activity

- OJPR coach splits members into groups of two and each group will perform the DG1 HVSS isolation procedure separately under the OJPR coach supervision.

- Once the OJPR coach is confident trainees can independently perform the isolation procedure, the group will move onto the next step.

OJPR Signature 6

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing HV equipment isolation procedure as an APC.

Activity 7: Performing HV Restoration Procedure as an APC

Demonstrate

- OJPR coach initiates activity by providing an overview of the restoration procedure and various restoration scenarios. The OJPR coach demonstrates while explaining how to restore the HV equipment IAW DG1 HVSS form

Practice Activity

- OJPR coach splits members into groups of two and each group will perform the DG1 HVSS restoration procedure separately under OJPR coach supervision

- Once the OJPR coach is confident trainees can independently perform the isolation procedure, the group will move onto the next step.

OJPR Signature 7

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing HV restoration procedure as an APC.

Activity 8: Performing PTW Procedure as an APC

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed PTW example IAW PC 002 scenario. The OJPR coach demonstrates while explaining how to perform the PTW procedure IAW PC 002 checklist

Practice Activity

- OJPR coach provides trainees with a blank PTW form and trainees complete the PTW procedure IAW PC 002 checklist individually while under OJPR coach supervision. Trainees then submit the completed form to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee

- Once the OJPR coach is confident trainees can independently perform the PTW procedure, the group will move onto the next step

OJPR Signature 8

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the PTW procedure as an APC

Activity 9: Performing SFT Procedure as an APC

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed SFT example IAW PC 002 scenario. The OJPR coach demonstrates while explaining how to perform the SFT procedure IAW PC 002 checklist

Practice Activity

- OJPR coach provides trainees with a blank SFT form and trainees complete the SFT procedure IAW PC 002 checklist individually while under OJPR coach supervision. Trainees then submit the completed form to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee

- Once the OJPR coach is confident trainees can independently perform the SFT procedure, the group will move onto the next step

OJPR Signature 9

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC

Activity 10: Performing LOA Procedure as an APC

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed LOA example. The OJPR coach demonstrates while explaining how to perform a LOA procedure.

Practice Activity

- OJPR coach provides trainees with a blank LOA form and new LOA scenario. Trainees complete the LOA procedure individually while under OJPR coach supervision. Trainees then complete the form and submit to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee

- Once the OJPR coach is confident trainees can independently perform the LOA procedure, the group will move onto the next step.

OJPR Signature 10

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC

Page Break

Activity 11: Performing APC Handover Process – DP

Demonstrate

- OJPR coach initiates this activity by providing an overview of the APC handover certificate process. The OJPR coach leads the trainees to the main switchboard location to demonstrate while explaining how to visually inspect and record I/O, mode and condition of all breakers

Practice Activity

- OJPR coach provides a blank APC handover certificate form to all trainees. The trainees practice completing the APC handover certificate while under the supervision of the OJPR coach. Trainees complete the form and submit to the OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee

- Once the OJPR coach is confident all trainees can complete the APC handover process form independently, the group will move onto the next step

OJPR Signature 11

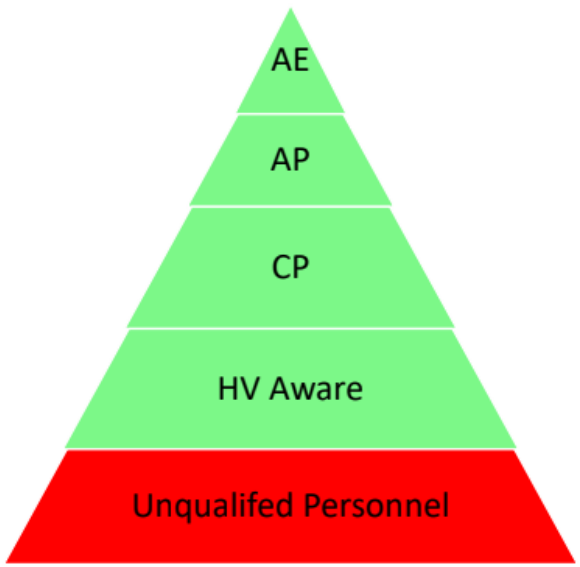
- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the SFGT procedure as an APC

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym / Abbreviation	Term or Definition
AE	Authorized Engineer
AP	Authorized Person
APC	Authorized Person in Control
CME	Circuit Main Earth
CP	Competent Person
HV	High Voltage
HVSP	High Voltage Switching Program
HVSS	High Voltage Switching Schedule
LOA	Limitation of Access
NTD	Naval Technical Department
PiC	Person in Charge
POW	Point of Work
PPE	Personal Protective Equipment
PTW	Permit to Work
SEMS	Safety and Environmental Management System
SFT	Sanction for Test
SSO	Ship’s Standing Orders
SSOW	Safe System of Work

5. HV Qualifications and Training Requirements

5.1 Qualification Levels



Authorizing Engineer (AE)
- Naval Technical Officer
- Senior Electrical Propulsion Manager

Authorized Person (AP)
- Leading Seaman and above Marine Technicians - Electrician

Competent Person (CP)
- All members of the Naval Technical Department, and other personnel where appropriate may also hold this qualification

HV Aware
- Entire ship's company, DND employees, regular contractors, and routine visitors

Unqualified Personnel
- All other visitors

Figure 5.1 – HV Qualification Hierarchy

		Distributed Learning	OJPR		
Authorized Person **4 days**	Stage 1 (30 mins)	Stage 2 (Unlimited)	Stage 4 (30 mins)	Stage 5 (145 mins)	
	Posted to AOPV Ship	PO 001 Complete AP instruction via DLN 3.0	AOPV HV Orientation	OJPR Signature: Demonstration and Practice Activity	
	• HV Training Onboarding Mtg • HV Training Requirements & Schedule • HV Training Student Guide	Lesson Objectives: L1: General introduction to HV L2: Electrical Hazards & Arc Flash L3: HV systems and compartments L4: HV states, stateboards, and pipes L5: HVSP locking devices L6: HV testing apparatus and types of testing L7: HV SSOW administration and system control L8: HV switching program procedures	Practice isolation DG1	Physical tour of HV compartments and machinery control room	1) Visually inspecting PPE 2) Donning arc flash PPE 3) Supervising PTW as a PiC AP 4) Supervising SFT as a PiC AP 5) Performing HVSP prep as an APC 6) Performing HV equipment isolation procedure as an APC 7) Performing HV restoration procedure as an APC 8) Performing PTW administration as an APC 9) Performing SFT

					administration as an APC 10) Performing LOA administration as an APC 11) Conducting APC certificate handover process

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1 para3-4, add any extra tables or paragraphs used for training resources in this annex. (Annex H not to be deleted)

ANNEX I – COSTING INFORMATION

Include information about the cost in whichever format is applicable (Not to be deleted).

Estimates only, based on CAF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

**5-year total based on one course per year, INCLUDING Initial Set-Up. Costs may be adjusted accordingly.

ANNEX J – COURSE HISTORY

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB (Dates)	(explanations of amendments, pertinent findings, trends, resource problems and modifications)

ANNEX K – ASSESSMENT PLAN

OJPR/ PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
OJPR	PO 001	Demonstrate proficiency in visually inspecting PPE	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in donning arc flash PPE	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in supervising PTW as a PiC AP	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in supervising SFT as a PiC AP	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in performing HVSP prep as an APC	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in performing HV equipment isolation procedures as an APC	OJPR signature	Pass	Retry until successful

OJPR	PO 001	Demonstrate proficiency in performing HV restoration procedures as an APC	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in performing PTW administration as an APC	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in performing SFT administration as an APC	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in performing LOA administration as an APC	OJPR signature	Pass	Retry until successful
OJPR	PO 001	Demonstrate proficiency in conducting APC certificate handover process	OJPR signature	Pass	Retry until successful
PC 002	PO 001	To assess the trainees' ability to perform HV work as an Authorized Person	Practical Assessment	80% & SAT on Critical Checklist items	Re-test/TRB upon 2nd unsuccessful PC

ANNEX L – TEL PLAN

PO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
PO 001	AP eLearning Course via DLN 3.0	F2F Classroom with PPTs	Impact on resources (availability of SMEs) to deliver F2F interim training
PO 001	DLN 3.0 (as required)	OJPR on ship	Liability on ships to conduct HV isolation procedure training in an uncontrolled state, including scheduling, disruptions and availability of ship as currently no alternative

MARPAC RDIMS # 1381603
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

ANNEX M – TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS

The following information was compiled during the QSPWB (date) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details - (Course name) candidates

“The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners” -CFITES Vol 4

Characteristic	Target population details
General data	
Potential group size (based on new AOPV ship): • AP: 10 (includes AE members) Expected annual throughput: 25 based on untrained full AOPV NTD crew Location(s) of the intended learners: Currently in HFX for the first two AOPV ships and then ESQ once the third AOPV ship arrives west coast. Access to computers (job): access to NFS computers and tablets. Gender: Male & Female Age: 19 to early 50s Trades: MSEO, MAR Tech Elec, MAR Tech Mech, WEng Tech	
Learning skills and preferences	
Competencies (general)	Trained Technician

Characteristic	Target population details
Reading and writing level	Strong
Preferences, attitudes toward a specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Open and flexible in using TEL
Educational level	University, College, High School
Previous military training experience	RQ-S1 qualified (4+ yrs)
Current subject-matter competence	
Experience with target performance	None
Ability to perform any part of the POs	Very High
Perception of the impact of achieving the POs on self, work or career	Excellent
Job/career stage	Early to late-career (4+ yrs)
Social Characteristics	
Relationship to peers	Excellent
Tendencies toward cooperation or competition	Cooperation